

MarineMamba: Multi-Granularity CLIP Feature Integration with Dual Mamba for Underwater Classification

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Abstract. Recognizing marine and aquatic species in underwater imagery is a fine-grained problem: species differ by subtle shape, fin, and texture cues, while images suffer from wavelength-dependent attenuation, scattering, and non-uniform illumination. Fully fine-tuning large CNN, Transformer, or state-space backbones on small, long-tailed collections can therefore be brittle. We propose MarineMamba, a lightweight classifier that keeps CLIP ViT-B/16 and ViT-B/32 frozen and trains only a compact dual-branch state-space head. Each branch projects the corresponding spatial grid, injects the global CLIP feature vector into the token sequence, scans patches in a locality-preserving bidirectional spiral order with a shared Mamba layer, and applies channel-and-spatial attention before fine and coarse descriptors are fused. Under the same training protocol, a fine-tuned ImageNet-pretrained Vim-tiny baseline obtains 89.14%, 89.58%, and 99.80% top-1 accuracy on AQUA20, Sea Animals 23, and Fish4Knowledge, respectively; MarineMamba reaches 93.46%, 95.90%, and 98.25% with 1.03M trainable parameters versus 6.96M. Ablations show that removing Mamba or feature-vector injection reduces AQUA20 accuracy by 7.46 and 8.37 points, identifying global-to-spatial propagation as the central mechanism. Overall, MarineMamba enables compact underwater recognition without full backbone fine-tuning. Our implementation is available at [Anonymized MarineMamba Source Code](#).

Keywords: Computer vision · Underwater recognition · CLIP · Mamba · State-space models

1 Introduction

Automated recognition of aquatic and marine species is central to biodiversity surveys, fisheries stock assessment, aquaculture monitoring, and coral-reef health assessment [7]. Manual annotation of underwater video and still imagery does not scale to modern camera-trap and autonomous-vehicle deployments, but underwater recognition is not a direct copy of natural-image classification. Species are often separated by subtle body-shape, fin, texture, or color-pattern differences; images suffer from wavelength-dependent color attenuation, scattering, blur, and non-uniform illumination; and field collections are naturally long-tailed.

These properties make full backbone fine-tuning a fragile default. Marine datasets are usually smaller and more imbalanced than the pretraining corpora

used for modern CNN, Transformer, or state-space backbones [4, 8, 10], so updating all backbone weights can overfit to a particular deployment, split, or species list. A complementary strategy is to freeze a broadly pretrained visual encoder and learn only a small, domain-specific head. CLIP [21] is well suited to this regime because its image encoder provides transferable global and spatial features, while CLIP ViT-B/16 and ViT-B/32¹ expose two granularities of evidence from the same image. The remaining question is how to use the spatial grids rather than only the pooled feature vectors.

State-space models provide a useful mechanism for this head because they propagate information through a linear-time recurrence. Mamba [8] and visual variants such as Vim [30], VMamba [15], and LocalMamba [11] have made selective state-space modeling a growing direction in vision [29]. These models also show that token ordering is not a minor implementation choice: a sequential model can only process a two-dimensional image grid after the grid has been serialized. Raster order creates artificial discontinuities at row boundaries, and those discontinuities are especially visible on the small CLIP feature grids used here. This motivates a spatial head that keeps the CLIP encoders frozen, preserves local adjacency during scanning, and fuses fine- and coarse-grained evidence without training a large backbone.

We aim to address underwater and aquatic species classification with a state-space recognition model. Although Mamba reduces sequence-modeling complexity relative to self-attention and can be lighter than many Transformer or CNN backbones, using it as a trainable visual backbone still requires updating millions of parameters, which is costly and prone to overfitting on small, imbalanced marine datasets. This motivates us to use pretrained CLIP as a frozen visual representation and learn only a compact state-space reader on top of it. MarineMamba therefore extracts frozen CLIP ViT-B/16 and ViT-B/32 spatial grids and feature vectors, projects each granularity into a compact hidden space, prepends the corresponding CLIP feature vector to the spatial tokens, and processes the sequence with a bidirectional spiral-scan Mamba block. After restoring the spatial layout, channel-and-spatial attention refines each branch; the two descriptors are concatenated and classified. Because the CLIP encoders are never updated, MarineMamba trains only 1.03M parameters, making it more suitable for resource-constrained deployment. We also explore fine-tuning Vim-tiny on the marine datasets under the same training setting, which yields strong results but requires substantially more trainable parameters.

The main contributions are:

- a lightweight frozen-CLIP recognition pipeline for underwater and aquatic species classification, using ViT-B/16 and ViT-B/32 grids as complementary granularities;
- a MarineMamba block that combines feature-vector injection, bidirectional spiral scanning, Mamba sequence modeling, and channel-and-spatial attention for small CLIP feature grids;

¹ OpenCLIP (open-source CLIP implementation)

- Experiments on AQUA20, Sea Animals 23, and Fish4Knowledge demonstrate superior top-1 accuracy while requiring significantly fewer parameters.

2 Related Work

Marine image recognition, situated within the broader underwater computer vision landscape [7], has followed the movement from hand-crafted features to fully learned backbones. CNNs such as ResNet [10], EfficientNet [23], and ConvNeXt [16] provide hierarchical local features, while Vision Transformers [4] replace convolutional aggregation with global self-attention. In underwater settings, these architectures must operate under color attenuation, scattering, background clutter, fine-grained species similarity, and uneven class frequencies. These factors make the evaluation protocol especially important: accuracy on a curated or balanced split may not reflect performance on naturally collected data [7].

Fish4Knowledge [5] is a representative benchmark for fish-species recognition. MLR-VGGNet [19] reports up to 97.09% test accuracy on Fish4Knowledge under a curated, class-balanced split, while ConvFishNet [20], inspired by FishNet [22], reports high precision on a natural split with substantially reduced computational cost. These methods demonstrate the effectiveness of specialized CNN designs for marine imagery, but their reported numbers are difficult to compare directly with raw long-tailed splits because preprocessing, partitions, and class distributions differ. More recent datasets and studies broaden this setting: AQUA20 [6] provides a challenging aquatic species benchmark, and AquaSelect [3] studies selective prediction so a trained classifier can abstain when confidence is low. These works highlight complementary problems: stronger marine representations, reliable confidence, and robustness to imbalance.

Visual recognition has advanced from convolutional neural networks [10, 23], which capture global context only weakly, to Vision Transformers [4], which model it globally via self-attention but at quadratic cost. State-space models offer a different route to global context, and recent surveys identify visual recognition as a central application area for Mamba-style architectures [29]. S4 [9] and Mamba [8] model long sequences through linear-time recurrence rather than quadratic self-attention. Vision adaptations therefore need a rule for serializing image grids. Vim [30] applies bidirectional state-space modeling to visual tokens, VMamba [15] aggregates multiple directional scans, LocalMamba [11] emphasizes local windows, and PlainMamba [28] studies non-hierarchical visual Mamba models. Together, these works make scan order a core modeling choice rather than a preprocessing detail.

Several Mamba studies are especially relevant to spatial ordering and feature fusion. SpiralMamba [24] and LSFMamba [2] show that spiral traversal can be useful when local continuity matters, particularly in hyperspectral classification. Closest in spirit to CLIP-guided state-space modeling, M³amba [1] couples CLIP with a Mamba fusion module for multi-modal remote sensing, but it targets sensor fusion rather than marine RGB species recognition.

Existing marine-recognition methods mostly rely on fully fine-tuned CNN or Transformer backbones, often under dataset protocols that differ in split construction and imbalance severity [7]. Existing visual Mamba methods, in turn, focus mainly on generic recognition, hyperspectral data, remote-sensing fusion, or modality-specific adaptation [29], and they do not directly address frozen multi-granularity CLIP feature grids for long-tailed underwater species classification. MarineMamba enters this gap by keeping the CLIP encoders frozen, using a small trainable state-space head, and scanning the fine and coarse CLIP spatial grids with locality-preserving spiral order before fusing them for marine data classification.

3 Methodology

MarineMamba separates visual representation learning from task-specific classification. Two frozen CLIP visual encoders provide complementary spatial features, and a compact state-space classifier — the only trainable component — models their fine- and coarse-scale structure. The fine branch operates on local patch evidence from CLIP ViT-B/16, the coarse branch on the broader context of CLIP ViT-B/32. Each branch injects the corresponding CLIP feature vector into a spiral-ordered Mamba sequence, and the two branch descriptors are concatenated before classification. Figure 1 gives an illustration.

Throughout, $r \in \{16, 32\}$ indexes the two granularities, $H_r \times W_r$ is the extent of the corresponding CLIP feature grid (14×14 and 7×7), $N_r = H_r W_r$, $d = 128$ is the shared hidden width, and C is the number of species. We write $[\cdot; \cdot]$ for concatenation along the token dimension and $\otimes$ for the element-wise multiplication.

3.1 Dual-Branch Encoder

Given an input image x , two frozen CLIP visual encoders Φ_{16} and Φ_{32} produce a spatial feature grid together with a global embedding, $(\mathbf{S}_r, \mathbf{g}_r) = \Phi_r(x)$, with

$$\mathbf{S}_r \in \mathbb{R}^{768 \times H_r \times W_r}, \quad \mathbf{g}_r \in \mathbb{R}^{512}, \quad r \in \{16, 32\}.$$

Neither encoder is updated during training; only the classifier below is optimized. This keeps the trainable parameter count small while preserving strong generic visual features, and it allows both grids to be extracted once and cached.

For each resolution, the spatial grid is flattened into N_r patch tokens, $\text{flat}(\mathbf{S}_r) \in \mathbb{R}^{N_r \times 768}$, projected to width d , and prefixed with the projected CLIP feature vector:

$$\mathbf{T}_r^0 = [\mathbf{W}_g^r \mathbf{g}_r; \text{flat}(\mathbf{S}_r) (\mathbf{W}_s^r)^\top] + \mathbf{E}_r \in \mathbb{R}^{(N_r+1) \times d}, \quad (1)$$

where $\mathbf{W}_g^r \in \mathbb{R}^{d \times 512}$ and $\mathbf{W}_s^r \in \mathbb{R}^{d \times 768}$ are learnable projections and $\mathbf{E}_r \in \mathbb{R}^{(N_r+1) \times d}$ is a learnable positional embedding. The fine branch therefore carries $1 + 14 \times 14 = 197$ tokens and the coarse branch $1 + 7 \times 7 = 50$.

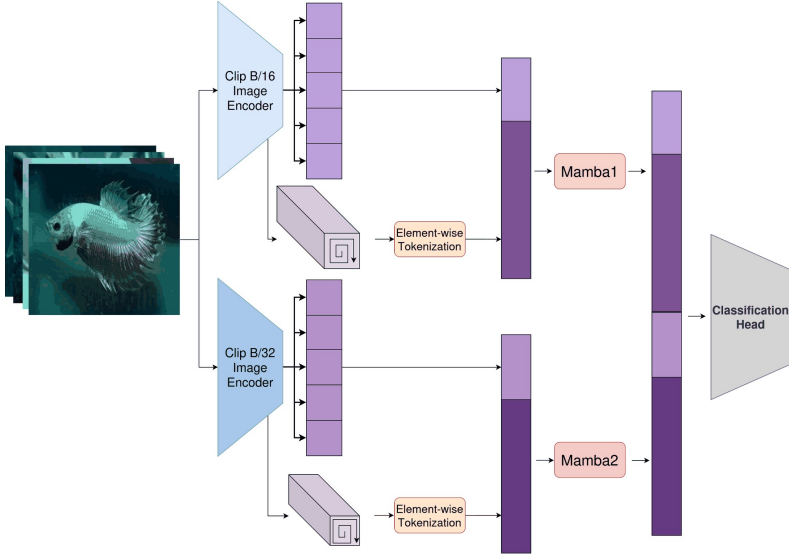


Fig. 1: Illustration of MarineMamba.

The sequence is refined by L MarineMamba blocks (Section 3.3), $\mathbf{T}_r^\ell = \mathcal{B}_r(\mathbf{T}_r^{\ell-1})$ for $\ell = 1, \dots, L$. The branch descriptor is the mean of the layer-normalized output over the injected CLIP feature vector and all spatial tokens,

$$\mathbf{h}_r = \frac{1}{N_r + 1} \sum_{j=0}^{N_r} \text{LN}(\mathbf{T}_r^L)_j \in \mathbb{R}^d. \quad (2)$$

3.2 Spiral Token Order

Let $\mathcal{G}_r = \{1, \dots, H_r\} \times \{1, \dots, W_r\}$ be the set of grid cells and let $c : \{1, \dots, N_r\} \rightarrow \mathcal{G}_r$ map a token index to its cell. A *scan order* is a bijection $\pi : \{1, \dots, N_r\} \rightarrow \{1, \dots, N_r\}$; token $\pi(k)$ is presented to the sequence model at step k . Two cells are *adjacent* when their ℓ_1 distance is 1.

We define the forward spiral order π_r by peeling the grid ring by ring from the top-left cell toward the center. The reverse order is $\bar{\pi}_r(k) = \pi_r(N_r - k + 1)$. Both orders are represented by permutation matrices $\mathbf{P}_r, \bar{\mathbf{P}}_r \in \{0, 1\}^{N_r \times N_r}$, so reordering is $\mathbf{P}_r \mathbf{T}_{\text{sp}}$ and restoring the raster layout is $\mathbf{P}_r^\top$. Spiral orders have also been used in hyperspectral Mamba models [2, 24]; here the purpose is whole-image classification over compact frozen-CLIP grids.

The motivation is locality preservation. A spiral is a Hamiltonian path in the grid graph, so consecutive tokens are always 4-adjacent and the total traversal distance is $N_r - 1$. Raster order instead jumps from the end of one row to the beginning of the next. Since Mamba propagates information through a sequential recurrence, reducing these artificial discontinuities gives the state-space layer a more geometrically coherent input.

3.3 MarineMamba Block

Write the block input as $\mathbf{T} = [\mathbf{t}_{fv}; \mathbf{T}_{sp}]$, separating the CLIP feature vector from the N_r spatial tokens. The spatial tokens are reordered in both spiral directions, the feature vector is prepended to each ordering, and both sequences are layer-normalized and passed through a *single, shared* Mamba layer $\mathcal{M}$:

$$\begin{aligned}\mathbf{Y}_f &= \mathcal{M}(\text{LN}[\mathbf{t}_{fv}; \mathbf{P}_r \mathbf{T}_{sp}]), \\ \mathbf{Y}_b &= \mathcal{M}(\text{LN}[\mathbf{t}_{fv}; \bar{\mathbf{P}}_r \mathbf{T}_{sp}]),\end{aligned}\quad (3)$$

so the bidirectional scan aggregates information outside-in and center-out at no additional parameter cost. Because the feature vector occupies the first position of both causal scans through the shared layer, its output is invariant to scan direction: the two leading outputs coincide, and we take that single vector directly as $\mathbf{y}_{fv}$ rather than averaging two identical copies. The remaining spatial outputs are returned to raster order and combined with a learnable scalar λ :

$$\mathbf{Y}_{sp} = \lambda \mathbf{P}_r^\top \mathbf{Y}_{f,sp} + (1 - \lambda) \bar{\mathbf{P}}_r^\top \mathbf{Y}_{b,sp}, \quad (4)$$

where λ is a learnable scalar initialized to 0.5.

The fused tokens are reshaped to a $d \times H_r \times W_r$ map and refined by sequential channel and spatial attention [27], applied in that order:

$$\mathbf{Y}'_{sp} = A_c(\mathbf{Y}_{sp}) \otimes \mathbf{Y}_{sp}, \quad \hat{\mathbf{Y}}_{sp} = A_s(\mathbf{Y}'_{sp}) \otimes \mathbf{Y}'_{sp}. \quad (5)$$

The block closes with residual updates,

$$\tilde{\mathbf{T}} = \mathbf{T} + [\mathbf{y}_{fv}; \hat{\mathbf{Y}}_{sp}], \quad \mathcal{B}_r(\mathbf{T}) = \tilde{\mathbf{T}} + \text{FFN}(\text{LN}(\tilde{\mathbf{T}})), \quad (6)$$

where FFN is a two-layer MLP with expansion ratio 4. LayerScale [26] is used in the implementation for stable residual updates.

The branch descriptors are concatenated into $\mathbf{h} = [\mathbf{h}_{16}; \mathbf{h}_{32}] \in \mathbb{R}^{2d}$ and classified by a two-layer MLP with a hidden width of $4d$:

$$\mathbf{z} = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \text{LN}(\mathbf{h})) \in \mathbb{R}^C, \quad \mathbf{W}_1 \in \mathbb{R}^{4d \times 2d}, \mathbf{W}_2 \in \mathbb{R}^{C \times 4d}. \quad (7)$$

Biases are omitted for brevity, and dropout is applied to the hidden activation during training.

3.4 Objective Function

Underwater datasets combine visually similar categories with long-tailed label distributions. We therefore train with a *class-weighted focal loss*. Let $p_{t,i} = \text{softmax}(\mathbf{z}_i)_{y_i}$ be the probability assigned to the ground-truth class y_i of the i -th sample. Over a mini-batch of size B ,

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \alpha_{y_i} (1 - p_{t,i})^\gamma \log p_{t,i}, \quad \alpha_c = \frac{N}{C n_c}, \quad (8)$$

with focusing parameter $\gamma = 2$, where N is the number of training samples, n_c the count of class c , and the α_c are rescaled to unit mean so that the effective learning rate is unchanged. The modulating factor $(1 - p_{t,i})^\gamma$ down-weights already well-classified examples [14], while α_c is a plain inverse-frequency weight; we do not use the effective-number reweighting sometimes given the same name.

4 Experiments

4.1 Datasets

We evaluate on three aquatic species classification datasets that differ in release time, scale, class count, and class balance.

AQUA20 [6], released in 2026, contains 8,171 images of 20 aquatic species with a predefined 6,559/1,612 train/test split.

Sea Animals 23 [13], released on Kaggle in 2023, contains 13,711 images of 23 sea animal species collected as a flat, unsplit image directory. We generate an 80/20 stratified train/test split per class with a fixed seed.

Fish4Knowledge [5], publicly documented in 2016, contains 23 fish species with 27,370 images in total. Unlike the curated, class-balanced split used by prior work [19], we keep the raw distribution, construct an 80/20 stratified split with at least 5 test images per class, and obtain an 881:1 training-set imbalance ratio between the most and least frequent species. This makes our Fish4Knowledge evaluation substantially harder than balanced-split reports.

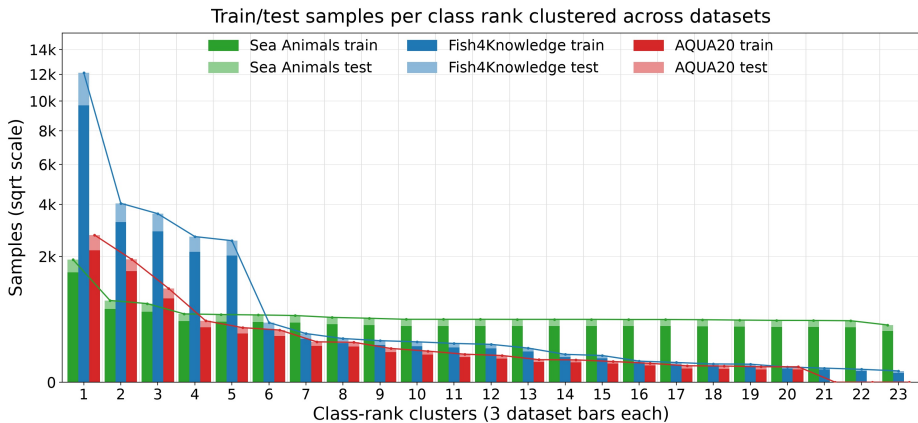
The class-frequency distributions exhibit a pronounced long tailed pattern, indicating substantial class imbalance across the evaluated datasets as shown in Figure 2. This imbalance is especially severe in Fish4Knowledge, where a small number of classes account for a disproportionately large fraction of samples while many classes remain sparsely represented. AQUA20 also shows a clear tailed distribution, although less extreme than Fish4Knowledge. In contrast, Sea Animals is comparatively more balanced, with class counts varying within a narrower range.

4.2 Implementation Details

Spatial CLIP features are extracted once with frozen ViT-B/16 (14×14 grid) and ViT-B/32 (7×7 grid) encoders and cached to disk. For the training split, each image contributes one clean view and three augmented views using random resized crop (scale 0.7–1.0), horizontal flip, RandAugment, and random erasing ($p = 0.25$); test features and global CLIP feature vectors use the clean CLIP preprocessing. Training then reads only the cached features.

Each branch uses hidden width $d = 128$ and one MarineMamba block. The Mamba layer uses $d_{\text{state}} = 32$, convolution width 4, and expansion 2; the classifier uses dropout 0.3. Batches use a square-root class-balanced sampler, and the objective is the class-weighted focal loss of Equation (8). We train MarineMamba and fine tune fVim-tiny (pretrained Vim-tiny [30] on Image-Net-1K)

Fig. 2: Class-wise train/test sample distribution showing long tailed imbalance, most prominently in Fish4Knowledge and AQUA20.



with AdamW, learning rate 3×10^{-4} , weight decay 0.05, batch size 128, 5 warmup epochs, cosine decay, and early stopping with 15-epoch patience on test top-1 accuracy. Unless noted otherwise, MarineMamba main results use 11 seeds ($\{0, \dots, 9, 42\}$), ablations use 3 seeds ($\{0, 1, 2\}$), and frozen-CLIP reference rows use the same schedule over 3 seeds. The fVim-tiny reference is fine tuned over 5 seeds ($\{0, 1, 2, 3, 4\}$). Single-valued entries are copied from cited sources and have no variance estimate.

4.3 Main Results

Table 1 reports fully fine-tuned CNN and Transformer results from the AQUA20 benchmark [6], with the ConvNeXt-Tiny and DeiT-Small rows from the concurrent AquaSelect study [3]. The frozen-CLIP references in Table 1 isolate the value of the spatial head. A linear probe on the two CLIP feature vectors reaches 81.97%, and a 0.27M-parameter MLP reaches 89.33%. MarineMamba reaches 93.46%, so the spatial-grid head contributes 4.13 points beyond a strong frozen-CLIP vector baseline. Compared with the selected fully fine-tuned backbones, MarineMamba outperforms the low-budget CNNs and ConvNeXt on top-1 while training only 1.03M parameters; it also exceeds our own fully fine-tuned fVim-tiny reference on this dataset despite far fewer trainable parameters.

Macro-F1 qualifies the AQUA20 result. MarineMamba reaches $83.38 \pm 1.95\%$, below ConvNeXt’s 88.92% and only 0.42 points above the frozen-CLIP MLP. As reported in Table 2 on Sea Animals 23, where the class distribution is less skewed, the pattern is stronger: MarineMamba reaches $95.90 \pm 0.80\%$ top-1 and $95.24 \pm 0.94\%$ macro-F1, exceeding both baselines.

Fish4Knowledge is the hardest imbalance setting. MarineMamba reaches $98.25 \pm 0.26\%$ top-1 and $87.14 \pm 1.33\%$ macro-F1 under an 881:1 training imbal-

Table 1: Selected results on AQUA20 (20 classes). Rows marked † are fully fine-tuned baselines reported by Fuad et al. [6]; ‡ denotes backbones fine-tuned by Daga et al. [3]. Parameter counts are trainable parameters.

Method	Params	Top-1 Acc.	Macro F1
<i>Frozen CLIP, no spatial head</i>			
CLIP zero-shot (ViT-B/16 prompts)	0	55.15%	52.94%
Linear probe on $[\mathbf{g}_{16}; \mathbf{g}_{32}]$	0.02 M	$81.97 \pm 0.06\%$	$67.65 \pm 0.54\%$
MLP on $[\mathbf{g}_{16}; \mathbf{g}_{32}]$ (CE)	0.27 M	$89.33 \pm 0.45\%$	$82.96 \pm 1.01\%$
MLP on $[\mathbf{g}_{16}; \mathbf{g}_{32}]$ (focal)	0.27 M	$85.24 \pm 0.13\%$	$79.88 \pm 0.54\%$
<i>Fully fine-tuned, comparable budget ($< 2M$)</i>			
† SqueezeNet1_1 [12]	0.73 M	76.80%	62.74%
† ShuffleNetV2 $\times 1.0$ [17]	1.27 M	78.66%	63.11%
<i>Fully fine-tuned, selected larger backbones</i>			
† ConvNeXt [16]	87.59 M	90.69%	88.92%
† ViT-B/16 [4]	85.80 M	86.29%	84.30%
‡ ConvNeXt-Tiny [16]	28.6 M	$87.63 \pm 0.41\%$	$79.91 \pm 1.88\%$
‡ DeiT-Small [25]	22.0 M	$85.98 \pm 1.16\%$	$77.32 \pm 1.86\%$
fVim-tiny	6.96 M	$89.14 \pm 0.29\%$	$82.47 \pm 1.85\%$
MarineMamba	1.03 M	$93.46 \pm 0.57\%$	$83.38 \pm 1.95\%$

Table 2: Results on Sea Animals 23. The ConvNeXt-Tiny row is reported by Daga et al. [3], who reserve 15% of the training split for validation. Our 80/20 stratified split yields a test set of comparable size but is not the identical partition.

Method	Params	Top-1 Acc.	Macro F1
ConvNeXt-Tiny [16] (Daga et al. [3])	28.6 M	$87.71 \pm 0.31\%$	$86.36 \pm 0.19\%$
fVim-tiny	6.96 M	$89.58 \pm 0.14\%$	$88.49 \pm 0.20\%$
MarineMamba	1.03 M	$95.90 \pm 0.80\%$	$95.24 \pm 0.94\%$

ance, while fVim-tiny achieves a remarkable 99.80% top-1 accuracy as listed in Table 3.

4.4 Embedding Visualization

To qualitatively examine how the trainable head reshapes the frozen representation, we project six embedding spaces to two dimensions with UMAP [18] (`n_neighbors`= 15, `min_dist`= 0.1, and a 50-component PCA pre-reduction), plotting train and test points together. The spaces are the two frozen CLIP spatial grids after mean pooling (768-d, ViT-B/16 and ViT-B/32), the fine and coarse MarineMamba branch descriptors (128-d each), their concatenation $\mathbf{h} = [\mathbf{h}_{16}; \mathbf{h}_{32}]$ (256-d, the vector the classifier actually receives), and a $64 \times 64 \times 3$ raw-pixel baseline. Figures 3, 4 and 5 show the result for AQUA20, Sea Animals 23, and Fish4Knowledge.

Table 3: Results on Fish4Knowledge (23 classes). The bottom block lists published results reported by Prasetyo et al.[§].

Method	Params	Top-1 Acc.	Macro F1
<i>Our controlled comparison — raw 881:1 split</i>			
fVim-tiny	6.96 M	99.80 ± 0.02%	98.86 ± 0.29% [†]
MarineMamba 1.03 M		98.25 ± 0.26%	87.14 ± 1.33%
<i>Published on a curated class-balanced split [19][§]</i>			
VGG16	14.71 M	82.44%	
VGG19	20.02 M	88.07%	
ResNet50	23.59 M	88.12%	context only,
Inception V3	21.80 M	85.66%	not directly
Xception	20.86 M	83.70%	comparable
MLR-VGG16	12.60 M	96.25%	
MLR-VGG19	15.55 M	97.09%	

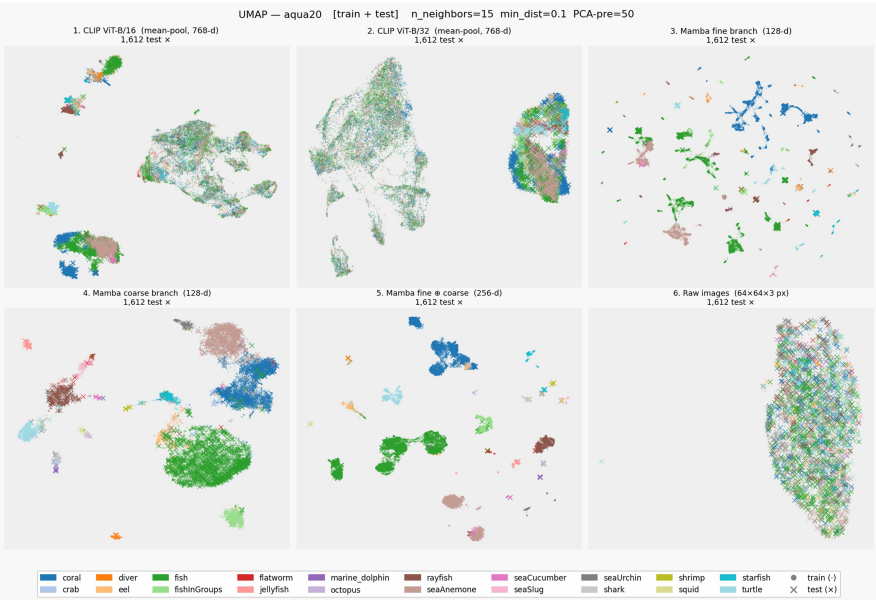


Fig. 3: UMAP projection of six embedding spaces on AQUA20.

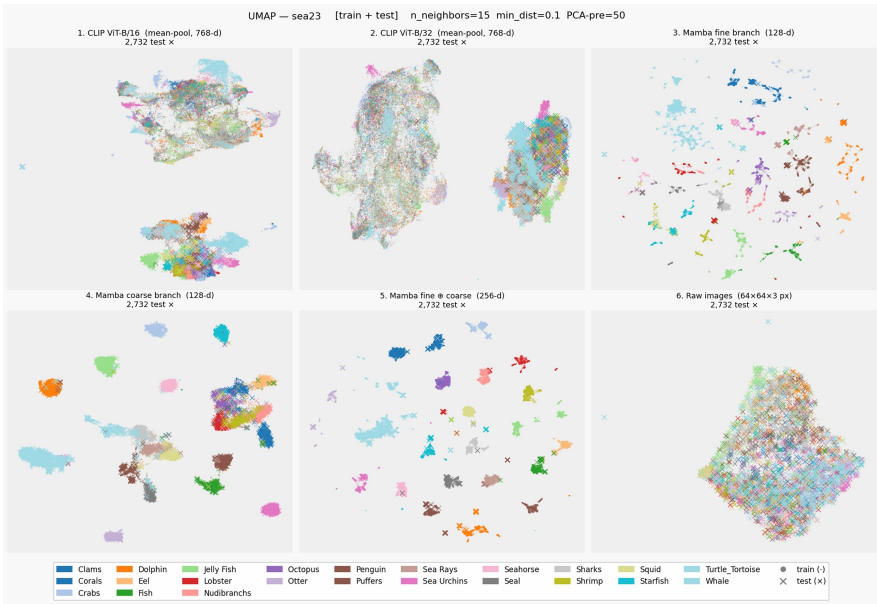


Fig. 4: UMAP projection of the same six embedding spaces on Sea Animals 23.

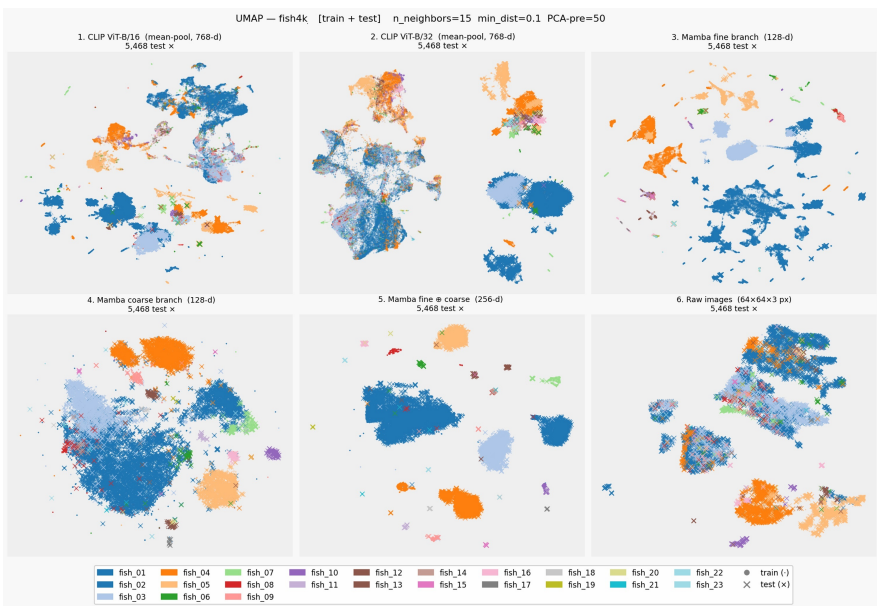


Fig. 5: UMAP projection of the same six embedding spaces on Fish4Knowledge.

4.5 Ablation Study

Table 4 reports AQUA20 ablations over 3 seeds. All variants consume the same cached CLIP features, so differences are attributable to the head.

Table 4: Ablation study on AQUA20.

Variant	Params	Top-1 Acc.	Δ vs. Full
Full model (spiral)	1.03 M	93.44 $\pm$ 0.13%	–
Raster scan order	1.03 M	92.91 $\pm$ 0.82%	–0.53%
Coarse branch only (B/32)	0.58 M	93.34 $\pm$ 0.51%	–0.10%
Fine branch only (B/16)	0.60 M	92.29 $\pm$ 0.93%	–1.15%
No focal loss (CE only)	1.03 M	92.83 $\pm$ 0.96%	–0.61%
No Mamba (token-wise FFN)	0.77 M	85.98 $\pm$ 0.44%	–7.46%
No feature-vector injection	1.03 M	85.07 $\pm$ 0.61%	–8.37%
<i>Reference</i> : MLP on [g ₁₆ ; g ₃₂], no spatial grid	0.27 M	89.33 $\pm$ 0.45%	–4.11%

The frozen-CLIP MLP reaches 89.33%, so the full head adds 4.11 points in this ablation setting. Removing the Mamba layer costs 7.46 points and removing feature-vector injection costs 8.37 points, with both variants falling below the spatial-grid-free MLP. This pattern indicates that the Mamba layer is valuable chiefly because it propagates the injected CLIP feature vector across spatial tokens before mean pooling; a token-wise FFN cannot perform that propagation.

Spiral scanning gives a modest mean gain over raster order (+0.53 points) but a large stability gain, reducing the standard deviation from 0.82% to 0.13%. The dual-branch design behaves similarly: the coarse-only branch nearly matches the full model’s mean (+0.10 points). Finally, focal loss improves this head by 0.61 points over cross-entropy.

5 Conclusion

This paper presented MarineMamba, a lightweight classifier that combines frozen CLIP ViT-B/16 and ViT-B/32 features with a spiral-scan Mamba head for underwater and aquatic species recognition. The method trains only 1.03M parameters, yet exceeds fVim-tiny and other baselines on AQUA20 and Sea Animals 23, while fVim-tiny has best results on highly imbalanced Fish4Knowledge split. The ablations clarify the mechanism: the main gain over a frozen-CLIP MLP comes from using the spatial grid, and the Mamba layer is most important for propagating the injected CLIP feature vector across spatial tokens before pooling.

Limitations. The results also set clear boundaries. MarineMamba’s AQUA20 macro-F1 trails ConvNeXt despite higher top-1 accuracy, indicating that its gains are stronger on majority classes than on the smallest classes. The dual-branch design and spiral order improve stability more clearly than mean accuracy

in isolation. On Fish4Knowledge, one non-rare class collapses to near-zero recall, and the retained fVim-tiny checkpoints are insufficient for a full per-class comparison. Future work will study class-specific failure diagnosis, additional CLIP granularities, and extensions to video or multi-view underwater data.

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